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Method to obtain low triterpene/triterpenoid-containing natural rubber latex

Abstract

The invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The invention further provides a laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, and an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

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Background/Summary

PRIORITY CLAIM

(1) This application claims priority to International Application No. PCT/EP2019/080668, filed Nov. 8, 2019, which claims priority to European Application No. 18205121.9, filed Nov. 8, 2018, wherein the contents of said applications are incorporated herein by reference in their entireties. Also, the entire contents of the ASCII text file entitled "IPM0119US Sequence Listing.txt" created on May 2, 2021, and having a size of 107 kilobytes, is incorporated herein by reference.

TECHNICAL FIELD OF THE INVENTION

(2) The present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated. Further, the present invention provides a laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. Additionally, the present invention also includes an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in said method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

BACKGROUND ART

(3) Supplied by the mevalonic acid (MVA) pathway oxidosqualene cyclases (OSCs) catalyse the cyclisation of 2,3-oxidosqualene (Abe, 2007; Thimmappa et al., 2014). This leads to the production of triterpenes, 30-carbon compounds, which comprise nearly 200 different scaffolds in plants (Xu et al., 2004). Tailoring enzymes can further modify triterpenes via acetylation, oxidation or by adding sugar chains, resulting in a huge diversity with more than 20,000 known triterpenoids (Xu et al., 2004; Thimmappa et al., 2014).

(4) Due to the cyclisation reaction type, a differentiation between two classes of triterpenoids exist: In the first class chair-boat-chair conformation occurs and results initially in the production of cycloartenol, which is a phytosterol intermediate (Abe, 2007; Thimmappa et al., 2014). The resulting sterols function e.g. as membrane compounds or signalling molecules. Therefore, they are important for plant development and growth (Hartmann, 1998; Clouse, 2002). In the second class chair-chair-chair conformation results among others in the formation of pentacyclic triterpenes such as lupeol and β -amyrin (Thimmappa et al., 2014). A large number of pentacyclic triterpenes show antimicrobial and antifungal characteristics and play a role in plant defence.

(5) Due to this characteristics, they are interesting for broad applications in food, cosmetics and pharmaceutical industries (Augustin et al., 2011; Osbourn et al., 2011; Moses et al., 2014). In dandelion, a high amount of pentacyclic triterpenes mainly occurs in roots, more precisely in laticifers, a specialised cell type, which produces a milky sap containing high concentrations of secondary metabolites (Furuno et al., 1993; Akashi et al., 1994). This cell type is also the place for poly(cis-1,4-isoprene) biosynthesis, the main component of natural rubber (NR) (Hagel et al., 2008). NR is an important biopolymer that is used for the production of more than 40,000 consumables, including e.g. tyres and medical devices (Davis, 1997; Mooibroek and Cornish, 2000; Hagel et al., 2008). High quality NR for commercial use usually derives from the rubber tree *Hevea brasiliensis*; however, it can also be extracted from the roots of the Russian dandelion, *T. koksaghyz*. NR derived from dandelion incorporates small amounts of additives like fatty acids, proteins and triterpenes, which have impact on the properties of NR (Nor and Ebdon, 1998; Sakdapipanich, 2007; Schulze Gronover et al., 2011).

(6) Since *T. koksaghyz* is a self-incompatible, sexually reproductive plant, it produces heterogenic offspring (Krotkov, 1945; Schulze Gronover et al., 2011). Consequently, the amount of pentacyclic triterpenes varies between single plants, which is not desirable for industrial applications.

(7) Moreover, Jungkong et al. and Ikeda et al. have shown that substances such as triterpenes generally have an important influence on the properties of natural rubber or latex.

(8) Accordingly, there is not only the need for a consistent level of triterpenes and/or triterpenoids, but also for a reduced content of triterpenes and/or triterpenoids of latex or rubber from laticiferous plants. The present invention aims to solve this problem.

SUMMARY OF THE INVENTION

(9) The problem is solved by the subject-matter as defined in the claims and as defined herein.

(10) In one embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The laticiferous plant or part thereof can also be a progeny of the respective laticiferous plant.

(11) In a preferred embodiment thereof, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

(12) In a further embodiment, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

(13) In one embodiment of the method of the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

(14) In a further embodiment of the method of the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant in this embodiment.

(15) In a specific embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

(16) In one embodiment of the method of the present invention, the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex, and mixtures thereof.

(17) The present invention also includes a laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

(18) In one embodiment, such a laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

(19) In another embodiment, in the laticiferous plant or part thereof of the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

(20) In a further embodiment, in the laticiferous plant or part thereof of the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

(21) In a further embodiment, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae, Moraceae and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group

consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybernum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a further embodiment, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae, Moraceae and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Cichorium intybus* L., *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybernum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a more preferred embodiment, the laticiferous plant or part thereof is *Cichorium intybus* L.

(22) In a further embodiment of the present invention, in the laticiferous plant or part thereof the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

(23) The present invention is also directed to a RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, preferably wherein the RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

(24) The invention will be better understood with reference to the detailed description when considered in conjunction with the drawings, the non-limiting examples and the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1: Analysis of TkOSC1- and TkOSC-RNAi lines. FIG. 1(a) shows normalised TkOSC1 expression of 9 to 16 weeks old wild-type (empty boxes, n=4), TkOSC1-RNAi (diagonally striped boxes, n=7) and TkOSC-RNAi plants (horizontally striped boxes, n=12) was determined by qRT-PCR using the constitutively expressed genes elongation factor 1a (TkEF1a) and ribosomal protein L27 (TkRP) from *T. koksaghyz*. FIG. 1(b) shows the expression analysis of TkOSC2-3, TkOSC5-6 and TkLUP, and FIG. 1(c) shows the one of TkSQS1 and TkSQE2. Therefore, three plants of each group were analysed. Quantification by GC/MS of single pentacyclic triterpenes shown in FIG.

1(d) and triterpene subclasses shown in FIG. 1(e) was performed on triterpene extracts from freeze-dried root material harvested from 5- to 7-month-old plants. Precursors=squalene and 2,3-oxidosqualene; sterol intermediates=cycloartenol and 24-methylene cycloartanol; pentacyclic triterpenes=taraxerol/n.i. (not identified pentacyclic triterpene), β -amyrin, lupeol, α -amyrin/lup-19(21)-en-3-ol, TkOSC3-product 1, TkOSC3-product 2, n.i.1 and taraxasterol; sterols=campesterol, stigmasterol and sitosterol; bar charts with appropriate standard deviations; boxplots: boxes=25%-75%, bar=median, square=average, hash=outlier, whisker=1.5 interquartile range; asterisks=statistical significance compared to *T. koksaghyz* wild-type (two-tailed t-test was applied for n=3, Mann-Whitney U test was applied for n>3, *=p<0.05, **=p<0.01).

(2) FIG. 2: Analysis of triterpenes in NR extracts from TkOSC1- and TkOSC-RNAi lines. NR was extracted from independent wild-type (empty boxes, n=3), TkOSC1-RNAi (diagonally striped boxes, n=3) and TkOSC-RNAi plants (horizontally striped boxes, n=3), the corresponding triterpene extracts were measured via GC/MS. Pentacyclic triterpenes (shown in FIG. 2(a)), triterpene subclasses (shown in FIG. 2(b)), total triterpenoids and total pentacyclic triterpenes (shown in FIG. 2(c)) were depicted in bar charts with appropriate standard deviations; asterisks indicate statistical significant differences between root and NR (two-tailed t-test, *=p<0.05, **=p<0.01, ***=p<0.001). Precursors=squalene and 2,3-oxidosqualene; sterol intermediates=cycloartenol and 24-methylene cycloartanol; pentacyclic triterpenes=taraxerol, β -amyrin, lupeol, α -amyrin, lup-19(21)-en-3-ol, taraxasterol, OSC3-products and not identified (n.i.) triterpenes; sterols=campesterol, stigmasterol and sitosterol; acetate=Ac.

(3) FIG. 3: Alignment of TkOSC1-8 and TkLUP amino acid sequences. The alignment was created with Clustal Omega (ebi.ac.uk/Tools/msa/clustalo/). Dashes represent gaps in aligned sequences. Conserved sites include a domain for product determination (M (W/Y) CY (C/S) R-motif, dashed box; e.g., SEQ ID NO: 6, amino acids 264-269), a region involved in substrate binding and cyclization (DCTAE-motif, solid box; e.g., SEQ ID NO: 8, amino acids 485-489) and repeated QW-motifs (underlined) involved in stabilization of carbocationic intermediates, which are localized near the 5' and 3'-end of the proteins. TkOSC1 relates to SEQ ID NO: 4, TKOSC2 relates to SEQ ID NO: 6, TKOSC3 relates to SEQ ID NO: 8, TkOSC4 relates to SEQ ID NO: 10, TkOSC5 relates to SEQ ID NO: 12, TKOSC6 relates to SEQ ID NO: 14, TkOSC7 relates to SEQ ID NO: 16, TKOSC8 relates to SEQ ID NO: 18 and TkLUP relates to SEQ ID NO: 20.

DETAILED DESCRIPTION OF THE INVENTION

(4) The invention is based on the surprising finding that the triterpene and/or triterpenoid content of latex or rubber can be reduced or even completely eliminated by the method according to the present invention. This allows the provision of a laticiferous plant, progeny thereof or part thereof, which is characterized by a low or even absent triterpene and/or triterpenoid content. This provides advantages for the products or materials, which are produced with rubber or latex received from these laticiferous plants, progeny thereof or parts thereof according to the present invention, for example with regard to durability, robustness and longevity.

(5) Example 2 shows that a knockdown of oxidosqualene cyclases (OSC) is possible by RNAi (RNA interference)-mediated knockdown. Example 3 shows that the knockdown of OSCs results in the reduction of the pentacyclic triterpenoid content (triterpenes and triterpenoids) in root and rubber of a laticiferous plant such as *Taraxacum koksaghyz*. For example, with regard to the root, the following observations were made: The pentacyclic triterpenoids were strongly affected in the transgenic lines (FIG. 1(d) and FIG. 1(e)). As a result, the total amount was reduced from 4.6 mg/g dry root weight in wild-type plants to 2.8 mg/g dry root weight in TkOSC1-RNAi plants comprising a nucleotide sequence of SEQ ID NO: 2 and to 2.6 mg/g in TkOSC-RNAi plants comprising a nucleotide sequence of SEQ ID NO: 1. Especially, the most abundant triterpenes, taraxasterol, β -amyrin and one unidentified triterpene (n.i.1), were remarkably reduced with a higher level of significance in TkOSC-RNAi plants. Besides the pentacyclic triterpenes, further compounds were measured via GC/MS (FIG. 1(e), Table 3). It was found that precursors like

squalene and 2,3-oxidosqualene accumulated significantly in the transgenic lines, with a higher level of significance in TkOSC-RNAi plants. Additionally, cycloartenol and 24-methylene cycloartanol, two sterol intermediates, were detected in both transgenic lines, with an average of 0.1 mg/g dry root weight in TkOSC1-RNAi plants and 0.9 mg/g dry root weight in TkOSC-RNAi plants. Sterols including campesterol, stigmasterol and sitosterol were detected to equal amounts of about 0.5 mg/g dry root weight in all plants and were not affected by the knock-down. In sum, the total triterpene content including precursors, sterol intermediates, sterols and pentacyclic triterpenes was significantly reduced in transgenic RNAi lines to about two-thirds with a higher level of significance in TkOSC-RNAi plants (FIG. 1(e)). In conclusion, it can be claimed that an OSC knockdown strongly reduces triterpenes in dandelion root and NR. For example, it could be observed that in dandelion root the total content of triterpenes could be reduced to a total content of 70%, while in dandelion NR the total content of triterpenes could be reduced to a total content of 44%.

(6) For example, with regard to the NR, the following observations were made: After rubber extraction from root material of wild-type, TkOSC1-RNAi and TkOSC-RNAi plants, the triterpenes in NR extracts were isolated by acetone extraction and quantified in relation to NR weight (FIG. 2, Table 4). The crude acetone extracts contained acetate derivatives of different triterpenes. Those acetates as well as the predominant triterpenes in latex (taraxasterol, β -amyirin and n.i.1) were strongly reduced compared to wild-type plants with a higher level of significance in TkOSC-RNAi plants compared to TkOSC1-RNAi plants (FIGS. 2(a) and 2(b)). The average amount of total pentacyclic triterpenes in wild-type plants with 56.6 mg/g NR was reduced to 33.4 mg/g NR in TkOSC1-RNAi lines or even to 16.1 mg/g NR in TkOSC-RNAi lines (FIG. 2(c)). In contrast to the reduced pentacyclic triterpene content, increasing amounts of triterpene precursors as well as sterol intermediates were again detected in RNAi-lines compared to wild-type, while equal amounts of sterols were measured in all three groups. However, total triterpenoid amount was still significantly decreased to two thirds in TkOSC1-RNAi NR and to less than the half in TkOSC-RNAi NR (FIG. 2(b)). In conclusion, it can be stated that an OSC knockdown strongly reduces triterpenes/triterpenoids in dandelion root and NR.

(7) The present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(8) In another embodiment, the present invention provides a method of reducing the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(9) In a further embodiment, the present invention provides a method of eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(10) In another embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(11) In a further embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex from a laticiferous plant or part thereof,

comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(12) In another embodiment, the present invention provides a method of reducing the content of triterpenes and/or triterpenoids of rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(13) In another embodiment, the present invention provides a method of reducing the content of triterpenes and/or triterpenoids of latex from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(14) In a further embodiment, the present invention provides a method of eliminating the content of triterpenes and/or triterpenoids of rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(15) In a further embodiment, the present invention provides a method of eliminating the content of triterpenes and/or triterpenoids of latex from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(16) In a further embodiment, the present invention provides a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant, progeny thereof or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(17) "Triterpenes" as used herein relate to a class of chemical compounds composed of three terpene units with the molecular formula $C_{30}H_{48}$; they may also be thought of as comprising of six isoprene units. Animals, plants and fungi all produce triterpenes, including squalene, the precursor to all steroids. Triterpenes as such do not comprise heteroatoms.

(18) "Triterpenoids" as used herein relate to triterpenes, which possess at least one heteroatom. Examples of triterpenoids include, but are not limited to, taraxerol, β -amyrin, lupeol, α -amyrin, lup-19(21)-en-3-ol or taraxasterol. Triterpenes and triterpenoids may also be summarized as pentacyclic triterpene content of the laticiferous plant or part thereof. Triterpenes and triterpenoids may also be summarized as pentacyclic triterpene content of the method of the present invention.

(19) As used herein, the term "progeny" or "progenies" refers in a non limiting manner to offspring or descendant plants. According to certain embodiments, the term "progeny" or "progenies" refers to plants developed or grown or produced from the respective laticiferous plants. Those plants grown under consideration of the methods according to the present invention as described herein are characterized by a reduced or eliminated content of triterpenes and/or triterpenoids of the latex or rubber, which is gained from these laticiferous plant progeny or progenies.

(20) For the method of the present invention, it is necessary that the activity of the oxidosqualene cyclase is reduced or eliminated compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof. There are many possible ways to achieve this reduction or elimination of oxidosqualene cyclase activity including reducing/eliminating the amount of the enzyme and/or reducing/eliminating the activity of the enzyme oxidosqualene cyclase. Ways to reduce the oxidosqualene cyclase amount, i.e. reduction of the expression of oxidosqualene cyclase, are known to a person skilled in the art and include the use gene knockdown technologies

such as RNA interference (RNAi) or modification of regulatory elements. The expression of the oxidosqualene cyclase can be eliminated e.g. by knocking out the gene(s) encoding an oxidosqualene cyclase. Examples of technologies for knocking out genes include homologous recombination or site-specific nucleases such as Zinc-Finger nucleases, TALEN or CRISPR.

(21) However, a “reduction of the content of triterpenes and/or triterpenoids” or “the content of triterpenes and/or triterpenoids is reduced” or “being capable of reducing the content of triterpenes and/or triterpenoids” means in the context of the present invention and as used herein that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

(22) An “elimination of the content of triterpenes and/or triterpenoids” or “the content of triterpenes and/or triterpenoids is eliminated” or “being capable of eliminating the content of triterpenes and/or triterpenoids” means in the context of the present invention and as used herein that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

(23) The “activity of at least one oxidosqualene cyclase is reduced” or the “reduction of the activity of the at least one oxidosqualene cyclase” or “being capable of reducing the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of the at least one oxidosqualene cyclase contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

(24) The “activity of at least one oxidosqualene cyclase is eliminated” or an “elimination of the activity of the at least one oxidosqualene cyclase” or “being capable of eliminating the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of at least one oxidosqualene cyclase of the laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

(25) “Latex” or “latexes” as used herein relate to products well known to those skilled in the art.

They consist of aqueous dispersions of water-insoluble polymers. These fluid systems contain, as dispersed phase, particles of polymers consisting of several entangled polymer chains in an aqueous dispersion medium. The diameter of the polymer particles within the dispersion can range between 5 nm and 10 μ m. These latexes find many applications, in particular as additives in formulations for paints, for paper (coating mixtures, bulk paper) or in formulations intended to be applied in the building industry (adhesive, bonding agents, smoothing coatings, etc.). They impart important properties on these formulations by virtue, for example, of their binding power and their film-forming power.

(26) “Rubber” as used herein relates to natural rubber, component of the latex of the rubber tree *Hevea brasiliensis* and of other rubber-supplying plants. It contains polyisoprene, which is present very finely dispersed in the aqueous phase and represents the actual natural rubber, and also, as secondary components, proteins, sugars, polysaccharides and inorganic salts. These secondary components, in particular, form a culture medium for bacteria, which multiply quickly in the latex and cause it to perish, which finally results in coagulation, i.e., in the precipitation of the polymer phase. There are often long distances, and possibly relatively long storage times, between extraction at the rubber plantations and further processing. If processing to solid rubber does not take place on site, the latex is normally treated with suitable agents immediately after extraction. This treatment is to provide protection against bacterial attack of the latex and for stabilization of the physico-chemical properties of the latex.

(27) The term “laticiferous” or “laticifers” as used in the context of the present invention and as used herein is related to or is a type of elongated secretory cells found in the leaves and/or stems of plants that produce latex and rubber as secondary metabolites. Laticifers may be divided into: Articulated laticifers, i.e., composed of a series of cells joined together, or non-articulated laticifers, consisting of one long coenocytic cell. Non-articulated laticifers begin their growth from the meristematic tissue of the embryo, termed the laticifer initial, and can exhibit continual growth throughout the lifetime of the plant. Laticifer tubes have irregularly edged walls and a larger inner diameter than the surrounding parenchyma cells. In the development of the cell, elongation occurs via karyokinesis and no cell plate develops resulting in coenocytic cells, which extend throughout the plant. These cells can reach up to tens of centimeters long and can be branched or unbranched. They are thought to have a role in wound healing and as defense against herbivory, as well as pathogen defense, and are often used for taxonomy.

(28) The term “wild-type” plant, respectively laticiferous plant, or “untreated” plant, respectively laticiferous plant, refers in the context of the present invention and as used herein to plants, which serve as starting material for the production of the plants described herein and according to the present invention, i.e. plants of which the genetic information—apart from the genetic modification introduced—is identical to the genetic information of the plant of the present invention. This specifically means herein that the “wild-type” plant, respectively laticiferous plant, or “untreated” plant, respectively laticiferous plant, is without being genetically modified or before being genetically modified as described herein.

(29) In an embodiment of the method of the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In one further embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “genetically modified” means, as used herein and in the context of the present invention, that the laticiferous plant, progeny thereof or part thereof comprises one or more heterologous nucleotide sequence(s) that is/are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. A

“heterologous nucleotide sequence” or “heterologous nucleic acid” is a sequence that is not naturally occurring in the respective nucleic acid. Generally, the heterologous nucleic acid comprises an open reading frame that encodes a polypeptide or non-translated RNA of interest (e.g., for delivery to a cell or subject). In another a further embodiment, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase. In a further embodiment, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase. Preferably, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. Also preferred, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. The term “is capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. Examples for such heterologous nucleotide sequences include nucleotides that can induce RNA interference, such as siRNA or shRNA.

(30) Accordingly, in one embodiment of the method of the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. Activity in this context may relate to a reduced or eliminated amount of triterpenes or triterpenoids in the resulting rubber or latex of the laticiferous plant, progeny thereof or part thereof. In this context “capable of reducing the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi. In this context, “capable of eliminating the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi.

(31) The amount or content of triterpenes or triterpenoids can be determined by methods known to a person skilled in the art. Such methods e.g. include, but are not limited to, mass spectrometry, chromatography such as RP-HPLC or gas chromatography, or NMR. An exemplary method for the quantification of triterpenes and/or triterpenoids in natural rubber is described in Example 1.

(32) In another embodiment of the method of the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase. Preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2, and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or

99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

(33) The nucleotide sequence according to SEQ ID NO: 1 of the present invention is given as follows:

(34) TABLE-US-00001

TCTCCAAGCGCCCATAGCGGAAAAGTGTTCTTATATTCTGCATAATGTAAC
ATGCAATTCTTCATGTACACTCCTGTTATTTCTGTTGAGGAAAGTCTCCA
TTATCCATTTGTCCATTTATGAGCAATTTTGCTGCCTTGTGTAATGGTGTC
GGATCTCTTTCAGCCTGTCCACCACATAGAAGACCAAGCATGGCCCATGAA
GTTTGCACAAAATTTGTTCTGTTCCCTTCCAAAGGTATAAATTTCTCTTGT
GGGCATGACTCAAAGTTCTCTCCCAACCACCTTCTGAGTTTTGTGTCGAG
AGGAAAAACTTGACAGCTTTCGAACTGCTTCACTATTTTCATATGTTTGT
CCACAAGCTACCAATCCTTGTAACACGAAATAGGTTCCATAAAGATAACAT
ATACCCCAATAACCATAACCAAGAACCATTTTCTTGT.

(35) The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 1 if the vector comprises a hairpin-structure, is given as follows:

(36) TABLE-US-00002 (SEQ ID NO: 47)

AACAAGAAAATGGTTCTTGGTATGGTTATTGGGGTATATGTTATCTTTATG
GAACCTATTTCTGTGTTACAAGGATTGGTAGCTTGTGGACAAACATATGAAA
ATAGTGAAGCAGTTCGGAAAGCTGTCAAGTTTTCTCTCGACACAAAAC
CAGAAGGTGGTTGGGGAGAGAAGCTTTGAGTCATGCCACAAAGAGAAATTA
TACCTTTGGAAGGGAACAGAACAAATTTTGTGCAAACCTTCATGGGCCATGC
TTGGTCTTCTATGTGGTGGACAGGCTGAAAGAGATCCGACACCATTACACA
AGGCAGCAAAATTGCTCATAAATGGACAAATGGATAATGGAGACTTTCCTC
AACAGGAAATAACAGGAGTGTACATGAAGAATTGCATGTTACATTATGCAG
AATATAGGAACACTTTTCCGCTATGGGCGCTTGGAGA.

(37) The nucleotide sequence according to SEQ ID NO: 2 of the present invention is given as follows:

(38) TABLE-US-00003

ATCCCAAGCTTGAATCGCACGGTTCAGTCTAACTGCCTTTCTCACGGCTGT
TGTCACCGCTTCATAGTTCACATCTTCGTCGTCTCCAAGTCTCACTGGAGG
TACGCTTATAAGATCAATTCCGC.

(39) The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 2 if the vector comprises a hairpin-structure, is given as follows:

(40) TABLE-US-00004 (SEQ ID NO: 48)

GCGGAATTGATCTTATAAGCGTACCTCCAGTGAGACTTGGAGACGACGAAG
ATGTGAACTATGAAGCGGTGACAACAGCCGTGAGAAAGGCAGTTAGACTGA
ACCGTGCGATTCAAGCTTGGGAT.

(41) More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

(42) Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

(43) More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%,

83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

(44) Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

(45) More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 1. More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 2, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 2.

(46) Two amino-acid or nucleotide sequences are said to be “identical” within the context of the present invention, if the sequence of amino-acid or nucleotide residues in the two sequences is the same when aligned for maximum correspondence as described below. Sequence comparisons between two (or more) peptides or polynucleotides are typically performed by comparing sequences of two optimally aligned sequences over a segment or “comparison window” to identify and compare local regions of sequence similarity. Optimal alignment of sequences for comparison may be conducted by the local homology algorithm of Smith and Waterman, *Ad. App. Math* 2: 482 (1981), by the homology alignment algorithm of Neddleman and Wunsch, *J. Mol. Biol.* 48: 443 (1970), by the search for similarity method of Pearson and Lipman, *Proc. Natl. Acad. Sci. (U.S.A.)* 85: 2444 (1988), by computerized implementation of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group (GCG), 575 Science Dr., Madison, Wis.), or by visual inspection.

(47) “Percentage of sequence identity” (or “degree” or “identity”) is determined by comparing two optimally aligned sequences over a comparison window, where the portion of the peptide or polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical amino-acid residue or nucleic acid base occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity.

(48) The definition of sequence identity given above is the definition that would use one of skill in the art. The definition by itself does not need the help of any algorithm, said algorithms being helpful only to achieve the optimal alignments of sequences, rather than the calculation of sequence identity.

(49) From the definition given above, it follows that there is a well-defined and only one value for the sequence identity between two compared sequences, which value corresponds to the value obtained for the best or optimal alignment.

(50) Instead of reducing the amount of the enzyme oxidosqualene cyclase, it is also within the context of the present invention that the reduction or elimination of the activity of the oxidosqualene cyclase may be inhibited by inhibitors. Accordingly, in an embodiment of the method of the present invention, the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. Further, in one embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. This means, that the term “genetically modified” or “genetically modification” may also comprise subjecting the laticiferous plant, progeny thereof

or part thereof of the present invention or the laticiferous plant, progeny thereof or part thereof used in any embodiments of the method of the present invention to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. Preferably, such an agent is an inhibitor of the activity of the at least one oxidosqualene cyclase. More preferably, the inhibitor of the activity of the at least one oxidosqualene cyclase is selected from the group consisting of 2,3-sulfidosqualene, derivatives of oxidosqualene, even more preferably 10- and 19-azasqualene derivatives, 4,4,10 β -trimethyl-trans-decal-3- β -ol (TMD) or derivatives thereof, aminoketones, azadecaline and derivatives of imidazoline, epohelmins, lanopylins, isoquinoline derivatives, 4-piperidinopyridines and 4-piperidinopyrimidines, even more preferably BIBB 515 (1-(4-chlorobenzoyl)-4-((4-2-oxazolin-2-yl)benzylidene))piperidine), vinyl sulfide and ketene dithioacetal derivatives, umbelliferone aminoalkyl derivatives and digalactosylglycerols.

(51) In another embodiment of the method of the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In another embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant, progeny thereof or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The above given definitions also apply for this embodiment, e.g. the term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above.

(52) In a further embodiment of the method of the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. In a further embodiment of the method of the present invention, the laticiferous plant, progeny thereof or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae.

(53) In a further embodiment of the method of the present invention, it is preferred that the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybema*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a further embodiment of the method of the present invention, it is preferred that the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Cichorium intybus* L., *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*,

Solidago fistulosa, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybemum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a more preferred embodiment, the laticiferous plant or part thereof is *Cichorium intybus* L. The term “crossings thereof” means in the context of the present invention plant varieties or lines that have been created by the deliberate interbreeding of closely or distantly related plant species, e.g. to introduce genes from one variety or line into a new genetic background. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant. It is even more preferred that the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysothamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof. However, most preferably the laticiferous plant or part thereof is *Taraxacum koksaghyz*. In one embodiment of the method of the present invention, the laticiferous plant or part thereof is not a member of the family Euphorbiaceae. This means the present invention is also directed to a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not a member of the family Euphorbiaceae. In another related embodiment of the method of the present invention, the laticiferous plant or part thereof is not *Hevea brasiliensis*. This means the present invention is also directed to a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not *Hevea brasiliensis*. In a further related embodiment of the method of the present invention, the laticiferous plant or part thereof is not from the genus *Landolphia*. This means, the present invention is also directed to a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not from the genus *Landolphia*.

(54) In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

(55) In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

(56) It is further preferred that the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8 and the oxidosqualene cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least

60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12.

(57) In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8 or 12.

(58) It is even more preferred that the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

(59) In another embodiment of the method of the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 4. It is most preferred that at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4.

(60) The nucleotide sequence of the oxidosqualene cyclase 1 is further given herein as SEQ ID NO: 3. The nucleotide sequence of the oxidosqualene cyclase 2 is further given herein as SEQ ID NO: 5. The nucleotide sequence of the oxidosqualene cyclase 3 is further given herein as SEQ ID NO: 7. The nucleotide sequence of the oxidosqualene cyclase 4 is further given herein as SEQ ID NO: 9. The nucleotide sequence of the oxidosqualene cyclase 5 is further given herein as SEQ ID NO: 11. The nucleotide sequence of the oxidosqualene cyclase 6 is further given herein as SEQ ID NO: 13. The nucleotide sequence of the oxidosqualene cyclase 7 is further given herein as SEQ ID NO: 15. The nucleotide sequence of the oxidosqualene cyclase 8 is further given herein as SEQ ID NO: 17. The nucleotide sequence of the lupeol synthase is further given herein as SEQ ID NO: 19.

(61) "Oxidosqualene cyclases (OSC)" as used herein are enzymes involved in the cyclization reactions of 2,3-oxidosqualene to form sterols or triterpenes. It can be differentiated between two different types of cyclisation reaction: The chair-boat-chair conformation results in the formation of cycloartenol, a phytosterol intermediate, while the chair-chair-chair conformation results in the formation of triterpenes (Abe, 2007; Thimmappa et al., 2014). The lupeol synthase also belongs to the oxidosqualene cyclases, wherein the lupeol synthase catalyses the cyclisation of 2,3-oxidosqualene to lupeol.

(62) The polypeptides of TkOSC1-8 and TkLUP according to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 and 20 of the present invention comprise 758 to 770 amino acids, whereas TkOSC1 compared to TkOSC2 displays the highest (94%) sequence identity among the newly identified sequences. Analysis of the amino acid sequence reveals that conserved domains reported for OSC are present with only slight modifications for TkOSC4, TkOSC5, TkOSC6 and TkOSC8. Those include a domain essential for product determination (M(W/Y)CY(C/S)R-motif), one region involved in substrate binding and polycyclization initiation (DCTAE-motif) and six repeated QW motifs involved in stabilization of carbocationic intermediates and localized near the 5' and 3'-ends of the proteins (see FIG. 3).

(63) In a further embodiment of the method of the present invention, the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex and mixtures thereof. In the Russian dandelion *Taraxacum koksaghyz*, the part of the laticiferous plant with the highest concentration of latex is the root. Accordingly, the part of the laticiferous plant preferably is root, if the plant or part thereof is *Taraxacum koksaghyz*. In the context of the present invention, the term "exudate(s)" comprises saps, gums, resins or nectar of the respective plant.

(64) The present invention also provides a laticiferous plant or part thereof being genetically

modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof. The present invention also provides a laticiferous plant, progeny thereof or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof.

(65) In one embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. In one embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above.

(66) The “activity of at least one oxidosqualene cyclase is reduced” or “reducing the activity of the at least one oxidosqualene cyclase” or “being capable of reducing the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of the at least one oxidosqualene cyclase contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

(67) The “activity of at least one oxidosqualene cyclase is eliminated” or “eliminating of the activity of the at least one oxidosqualene cyclase” or “being capable of eliminating the activity of at least one oxidosqualene cyclase” means in the context of the present invention and as used herein that the activity of at least one oxidosqualene cyclase of the laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

(68) In another embodiment, the laticiferous plant or part thereof according to the present invention, is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In another embodiment, the laticiferous plant, progeny thereof or part thereof according to the present invention, is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The term “is/are capable of reducing or eliminating” is also to be considered compared to the respective wild-type or untreated laticiferous plant, progeny thereof or part thereof as defined above. Such an agent may be an inhibitor of the activity of the at least one oxidosqualene cyclase, preferably an

inhibitor of the activity of the at least one oxidosqualene cyclase selected from the group consisting of 2,3-sulfidosqualene, derivatives of oxidosqualene, more preferably 10- and 19-azasqualene derivatives, 4,4,10 β -trimethyl-trans-decal-3- β -ol (TMD) or derivatives thereof, aminoketones, azadecaline and derivatives of imidazoline, epohelmins, lanopylins, isoquinoline derivatives, 4-piperidinopyridines and 4-piperidinopyrimidines, even more preferably BIBB 515 (1-(4-chlorobenzoyl)-4-((4-2-oxazolin-2-yl)benzylidene))piperidine), vinyl sulfide and ketene dithioacetal derivatives, umbelliferone aminoalkyl derivatives and digalactosylglycerols. In a further embodiment, such an agent may be an inhibitor of the activity of the at least one oxidosqualene cyclase and preferably an inhibitor of the activity of the at least one oxidosqualene cyclase selected from the group consisting of 2,3-sulfidosqualene, derivatives of oxidosqualene, more preferably 10- and 19-azasqualene derivatives, 4,4,10 β -trimethyl-trans-decal-3- β -ol (TMD) or derivatives thereof, aminoketones, ethyl methanesulfonates, azadecaline and derivatives of imidazoline, epohelmins, lanopylins, isoquinoline derivatives, 4-piperidinopyridines and 4-piperidinopyrimidines, even more preferably BIBB 515 (1-(4-chlorobenzoyl)-4-((4-2-oxazolin-2-yl)benzylidene))piperidine), vinyl sulfide and ketene dithioacetal derivatives, umbelliferone aminoalkyl derivatives and digalactosylglycerols. In a preferred embodiment, an inhibitor of the activity of the at least one oxidosqualene cyclase is ethyl methanesulfonates.

(69) In a further embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In a further embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the laticiferous plant, progeny thereof or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant, progeny thereof or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. The “activity of at least one oxidosqualene cyclase is reduced” or “reducing the activity of the at least one oxidosqualene cyclase” or “being capable of reducing the activity of at least one oxidosqualene cyclase” means in this context and as used herein that the activity of the at least one oxidosqualene cyclase contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified. The “activity of at least one oxidosqualene cyclase is eliminated” or “eliminating the activity of the at least one oxidosqualene cyclase” or “being capable of eliminating the activity of at least one oxidosqualene cyclase” means in this context that the activity of at least one oxidosqualene cyclase of the laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%,

99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified. (70) Thus, in one embodiment of the present invention, the laticiferous plant or part thereof may be a transgenic plant. In one embodiment of the present invention, the laticiferous plant, progeny thereof or part thereof may be a transgenic plant. A “transgenic plant” as used herein refers to a plant that contains genetic material not found in a wild-type plant of the same species, variety or cultivar. The genetic material may include a transgene, the product of a mutagenesis event (such as by transposon or T-DNA insertional mutagenesis), including insertion, deletion or substitution, an activation tagging sequence, a mutated sequence, the product of a homologous recombination event or a sequence modified by chimeraplasty. Typically, the foreign genetic material has been introduced into the plant by human manipulation, but any method can be used as one of skill in the art recognizes. A transgenic plant may contain an expression vector or cassette. The expression cassette typically comprises a polypeptide-encoding sequence operably linked (i.e., under regulatory control of) to appropriate inducible or constitutive regulatory sequences that allow for the expression of polypeptide. The expression cassette can be introduced into a plant by transformation or by breeding after transformation of a parent plant. The term “plant” as used herein refers to a whole plant, including seedlings and mature plants, as well as to a plant part, such as seed, fruit, leaf, or root, plant tissue, plant cells or any other plant material, e.g., a plant explant, as well as to progeny thereof, and to in vitro systems that mimic biochemical or cellular components or processes in a cell.

(71) In one embodiment of the laticiferous plant or part thereof according to the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In one embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase. In this context, “capable of reducing the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi. In this context, “capable of eliminating the activity of the at least one oxidosqualene cyclase” means that the respective RNAi can reduce the activity of the at least one oxidosqualene cyclase contained in the laticiferous plant, progeny thereof or part thereof for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00%, compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without or in the absence of the respective RNAi.

(72) In a further embodiment of the laticiferous plant or part thereof according to the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a

nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2. In a further embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

(73) Also for all embodiments of the laticiferous plant, progeny thereof or part thereof according to the present invention, the nucleotide sequence according to SEQ ID NO: 1 is given as follows:

(74) TABLE-US-00005

TCTCCAAGCGCCCATAGCGGAAAAGTGTTCTTCTGCATAATGTAAC
ATGCAATTCTTCATGTACACTCCTGTTATTTCTGTTGAGGAAAGTCTCCA
TTATCCATTTGTCCATTTATGAGCAATTTTGCTGCCTTGTGTAATGGTGTC
GGATCTCTTTCAGCCTGTCCACCACATAGAAGACCAAGCATGGCCCATGAA
GTTTGCACAAAATTTGTTCTGTTCCCTTCCAAAGGTATAAATTTCTCTTGT
GGGCATGACTCAAAGTTCTCTCCCAACCACCTTCTGAGTTTTGTGTCGAG
AGGAAAACTTGACAGCTTTCGAACTGCTTCACTATTTTCATATGTTTGT
CCACAAGCTACCAATCCTTGTAACACGAAATAGGTTCCATAAAGATAACAT
ATACCCCAATAACCATAACCAAGAACCATTTTCTTGTT.

(75) The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 1 if the vector comprises a hairpin-structure, is given as follows:

(76) TABLE-US-00006 (SEQ ID NO: 47)

AACAAGAAAATGGTTCTTGGTATGGTTATTGGGGTATATGTTATCTTTATG
GAACCTATTTCTGTGTTACAAGGATTGGTAGCTTGTGGACAAACATATGAAA
ATAGTGAAGCAGTTCGGAAAGCTGTCAAGTTTTTCTCTCGACACAAAAC
CAGAAGGTGGTTGGGGAGAGAACTTTGAGTCATGCCCACAAGAGAAATTA
TACCTTTGGAAGGGAACAGAACAAATTTTGTGCAAACTTCATGGGCCATGC
TTGGTCTTCTATGTGGTGGACAGGCTGAAAGAGATCCGACACCATTACACA
AGGCAGCAAAATTGCTCATAAATGGACAAATGGATAATGGAGACTTTCTC
AACAGGAAATAACAGGAGTGATCATGAAGAATTGCATGTTACATTATGCAG
AATATAGGAACACTTTTCCGCTATGGGCGCTTGGAGA.

(77) Also for all embodiments of the laticiferous plant, progeny thereof or part thereof according to the present invention, the nucleotide sequence according to SEQ ID NO: 2 of the present invention is given as follows:

(78) TABLE-US-00007

ATCCCAAGCTTGAATCGCACGGTTCAGTCTAACTGCCTTTCTCACGGCTGT
TGTCACCGCTTCATAGTTCACATCTTCGTCGTCTCCAAGTCTCACTGGAGG
TACGCTTATAAGATCAATTCCGC.

(79) The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 2 if the vector comprises a hairpin-structure, is given as follows:

(80) TABLE-US-00008 (SEQ ID NO: 48)

GCGGAATTGATCTTATAAGCGTACCTCCAGTGAGACTTGGAGACGACGAAG
ATGTGAACTATGAAGCGGTGACAACAGCCGTGAGAAAGGCAGTTAGACTGA
ACCGTGCGATTCAAGCTTGGGAT.

(81) More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%

or even 100% identical to SEQ ID NO: 1.

(82) Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

(83) More preferably, the at least one RNAi comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

(84) Even more preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

(85) More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 1. More preferably, the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 2, even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 2.

(86) In a further embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. Preferably, the laticiferous plant or part thereof according to the present invention is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybernum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a further embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae. Preferably, the laticiferous plant or part thereof according to the present invention is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Cichorium intybus* L., *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybemum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof. In a more preferred embodiment, the laticiferous plant or part thereof is *Cichorium intybus* L. It is even more preferred that the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysothamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof. Most preferably, the laticiferous plant or part thereof is *Taraxacum koksaghyz*. In one embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is not a member of the family Euphorbiaceae. This means the present invention also provides a laticiferous plant or part thereof being genetically

modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not a member of the family Euphorbiaceae. In another related embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is not *Hevea brasiliensis*. This means, the present invention also provides a laticiferous plant or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not *Hevea brasiliensis*. In a further related embodiment of the laticiferous plant or part thereof according to the present invention, the laticiferous plant or part thereof is not from the genus *Landolphia*. This means the present invention also provides a laticiferous plant or part thereof being genetically modified such that the activity of the at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof, wherein the laticiferous plant or part thereof is not from the genus *Landolphia*. The laticiferous plant or part thereof can also be the progeny of the respective laticiferous plant as mentioned in this paragraph.

(87) In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20. Preferably, the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12. More preferably, the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto. Most preferably, the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4.

(88) In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20.

(89) In another embodiment of the laticiferous plant or part thereof according to the present invention, it is further preferred that the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8 and the oxidosqualene cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12. In a further embodiment of the laticiferous plant, progeny thereof or part thereof according to the present invention, the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being

at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

(90) In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NOs: 4, 6, 8 or 12.

(91) It is even more preferred for the laticiferous plant or part thereof according to the present invention that the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

(92) In another embodiment of the laticiferous plant or part thereof according to the present invention, the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 4. It is most preferred that at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4.

(93) The nucleotide sequence of the oxidosqualene cyclase 1 is further given herein as SEQ ID NO: 3. The nucleotide sequence of the oxidosqualene cyclase 2 is further given herein as SEQ ID NO: 5. The nucleotide sequence of the oxidosqualene cyclase 3 is further given herein as SEQ ID NO: 7. The nucleotide sequence of the oxidosqualene cyclase 4 is further given herein as SEQ ID NO: 9. The nucleotide sequence of the oxidosqualene cyclase 5 is further given herein as SEQ ID NO: 11. The nucleotide sequence of the oxidosqualene cyclase 6 is further given herein as SEQ ID NO: 13. The nucleotide sequence of the oxidosqualene cyclase 7 is further given herein as SEQ ID NO: 15. The nucleotide sequence of the oxidosqualene cyclase 8 is further given herein as SEQ ID NO: 17. The nucleotide sequence of the lupeol synthase is further given herein as SEQ ID NO: 19.

(94) The definitions as well as the preferred embodiments provided herein above with regard to the method of the present invention apply mutatis mutandis also to the embodiments relating to the laticiferous plant, progeny thereof or part thereof of the present invention. Further, the definitions as well as the embodiments and preferred embodiments provided herein above with regard to the laticiferous plant or part thereof apply mutatis mutandis also to the progeny of the respective laticiferous plant or part thereof.

(95) The present invention also provides an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof. The present invention also provides an RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant, progeny thereof or part thereof. In this context, the term “reducing the content of triterpenes and/or triterpenoids” or the term “the content of triterpenes and/or triterpenoids is reduced” or the term “being capable of reducing the content of triterpenes and/or triterpenoids” means, as used herein, that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94% or 95% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically

modified or before being genetically modified. Further, the term “eliminating the content of triterpenes and/or triterpenoids” or the term “the content of triterpenes and/or triterpenoids is eliminated” or the term “being capable of eliminating the content of triterpenes and/or triterpenoids” means that the content of triterpenes and/or triterpenoids contained in a laticiferous plant, progeny thereof or part thereof is reduced for at least 95.1%, 95.2%, 95.3%, 95.4%, 95.5%, 95.6%, 95.7%, 95.8%, 95.9%, 96.0%, 96.1%, 96.2%, 96.3%, 96.4%, 96.5%, 96.6%, 96.7%, 96.8%, 96.9%, 97.0%, 97.1%, 97.2%, 97.3%, 97.4%, 97.5%, 97.6%, 97.7%, 97.8%, 97.9%, 98.0%, 98.1%, 98.2%, 98.3%, 98.4%, 98.5%, 98.6%, 98.7%, 98.8%, 98.9%, 99.0%, 99.1%, 99.2%, 99.3%, 99.4%, 99.5%, 99.6%, 99.7%, 99.8%, 99.9% or even 100.00% compared to the wild-type or untreated laticiferous plant, progeny thereof or part thereof, meaning without being genetically modified or before being genetically modified.

(96) Preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof. The laticiferous plant or part thereof can also be a progeny of the respective laticiferous plant as mentioned herein.

(97) In a further embodiment of the present invention, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1. More preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 1 and/or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1. Most preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 1.

(98) In a further embodiment of the present invention, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 2. More preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 2 and/or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 2. Most preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 2.

(99) Also for all embodiments of the RNAi according to the present invention, the nucleotide sequence according to SEQ ID NO: 1 is given as follows:

(100) TABLE-US-00009

TCTCCAAGCGCCCATAGCGGAAAAGTGTTCCCTATATTCTGCATAATGTAAC
ATGCAATTCTTCATGTACACTCCTGTTATTTCTGTTGAGGAAAGTCTCCA
TTATCCATTTGTCCATTTATGAGCAATTTTGCTGCCTTGTGTAATGGTGTC
GGATCTCTTTCAGCCTGTCCACCACATAGAAGACCAAGCATGGCCCATGAA
GTTTGCACAAAATTTGTTCTGTTCCCTTCCAAAGGTATAAATTTCTCTTGT
GGGCATGACTCAAAGTTCTCTCCCAACCACCTTCTGAGTTTTGTGTCGAG
AGGAAAACTTGACAGCTTTCGAACTGCTTCACTATTTTCATATGTTTGT
CCACAAGCTACCAATCCTTGTAACACGAAATAGGTTCCATAAAGATAACAT
ATACCCCAATAACCATACCAAGAACCATTTTCTTGTT.

(101) The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 1 if the vector comprises a hairpin-structure, is given as follows:

(102) TABLE-US-00010 (SEQ ID NO: 47)

AACAAGAAAATGGTTCTTGGTATGGTTATTGGGGTATATGTTATCTTTATG
GAACCTATTTCTGTGTTACAAGGATTGGTAGCTTGTGGACAAACATATGAAA

ATAGTGAAGCTTCGGTCAAGTTTTTCCTCTCGACACAAAAC
CAGAAGGTGGTTGGGGAGAGAACTTTGAGTCATGCCCCACAAGAGAAATTTA
TACCTTTGGAAGGGAACAGAACAAATTTTGTGCAAACCTTCATGGGCCATGC
TTGGTCTTCTATGTGGTGGACAGGCTGAAAGAGATCCGACACCATTACACA
AGGCAGCAAAATTGCTCATAAATGGACAAATGGATAATGGAGACTTTCCTC
AACAGGAAATAACAGGAGTGTACATGAAGAATTGCATGTTACATTATGCAG
AATATAGGAACACTTTTCCGCTATGGGCGCTTGGAGA.

(103) Also for all embodiments of the RNAi according to the present invention, the nucleotide sequence according to SEQ ID NO: 2 of the present invention is given as follows:

(104) TABLE-US-00011

ATCCCAAGCTTGAATCGCACGGTTCAGTCTAACTGCCTTTCTCACGGCTGT
TGTCACCGCTTCATAGTTCACATCTTCGTCGTCTCCAAGTCTCACTGGAGG
TACGCTTATAAGATCAATTCCGC.

(105) The complementary sequence thereto, which is, for example, also present in a vector construct comprising the SEQ ID NO: 2 if the vector comprises a hairpin-structure, is given as follows:

(106) TABLE-US-00012 (SEQ ID NO: 48)

GCGGAATTGATCTTATAAGCGTACCTCCAGTGAGACTTGGAGACGACGAAG
ATGTGAACTATGAAGCGGTGACAACAGCCGTGAGAAAGGCAGTTAGACTGA
ACCGTGCGATTCAAGCTTGGGAT.

(107) Preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

(108) More preferably, the at least one RNAi consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 1.

(109) Preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

(110) More preferably, the RNAi according to the present invention consists of a nucleotide sequence being at least 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or even 100% identical to SEQ ID NO: 2.

(111) More preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 1. Even more preferably, the RNAi according to the present invention consists of a nucleotide sequence according to SEQ ID NO: 1.

(112) More preferably, the RNAi according to the present invention comprises or consists of a nucleotide sequence according to SEQ ID NO: 2. Even more preferably, the at least one RNAi consists of a nucleotide sequence according to SEQ ID NO: 2.

(113) The term “RNA interference” refers generally and in the context of the present invention as used herein to a process in which a double-stranded RNA molecule changes the expression of a nucleic acid sequence with which the double-stranded or short hairpin RNA molecule shares substantial or total homology.

(114) The term “RNAi” refers generally and in the context of the present invention as used herein to an RNA sequence that elicits RNA interference, and which is transcribed from a vector. The terms “short hairpin RNA” or “shRNA” refer to an RNA structure having a duplex region and a loop region. This term should also be understood to specifically include RNA molecules with stem-loop or panhandle secondary structures. In some embodiments of the present invention, RNAi's are expressed initially as shRNAs.

(115) A better understanding of the present invention and of its advantages will be available from the following examples, offered for illustrative purposes only. The examples are not intended to limit the scope of the present invention in any way.

(116) The invention is further characterized by the following items:

(117) 1. Method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(118) 2. Method according to item 1, wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

(119) 3. Method according to item 2, wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

(120) 4. Method according to item 3, wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

(121) 5. Method according to any one of the preceding items, wherein the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae.

(122) 6. Method according to any one of the preceding items, wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybemum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysothamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof, more preferably wherein the laticiferous plant or part thereof is *Taraxacum koksaghyz*.

(123) 7. Method according to any one of the preceding items, wherein the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10,

the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20, preferably wherein the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8 and the oxidosqualene cyclase 5 according to SEQ ID NO: 12 and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12, more preferably wherein the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

(124) 8. Method according to any one of the preceding items, wherein the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex and mixtures thereof.

(125) 9. Laticiferous plant or part thereof being genetically modified such that the activity of at least one oxidosqualene cyclase is reduced or eliminated, compared to the respective wild-type or untreated laticiferous plant or part thereof.

(126) 10. Laticiferous plant or part thereof according to item 9, wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase, preferably that is capable of reducing or eliminating the expression of the gene of the at least one oxidosqualene cyclase, and/or wherein the laticiferous plant or part thereof is genetically modified by subjecting it to and/or incubating it with agents, which are capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

(127) 11. Laticiferous plant or part thereof according to item 9 or 10, wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase with the proviso that the at least one RNAi is capable of reducing or eliminating the activity of the at least one oxidosqualene cyclase.

(128) 12. Laticiferous plant or part thereof according to item 11, wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase, preferably wherein the at least one RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2.

(129) 13. Laticiferous plant or part thereof according to any one of items 9 to 12, wherein the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicomiculatum*, *Taraxacum hybemum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof, preferably wherein the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysothamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof, more preferably wherein the laticiferous plant or part thereof is *Taraxacum koksaghyz*.

(130) 14. Laticiferous plant or part thereof according to any one of items 9 to 13, wherein the at

least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, the lupeol synthase according to SEQ ID NO: 20, and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18 or 20, preferably wherein the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, and an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NOs: 4, 6, 8 or 12, more preferably wherein the at least one oxidosqualene cyclase is the oxidosqualene cyclase 1 according to SEQ ID NO: 4 or an amino acid sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical thereto.

(131) 15. RNAi being specifically directed against a mRNA transcript of at least one oxidosqualene cyclase, preferably wherein the RNAi comprises or consists of a nucleotide sequence according to SEQ ID NO: 1 and/or comprises or consists of a nucleotide sequence according to SEQ ID NO: 2 and/or comprises or consists of a nucleotide sequence being at least 60%, 70%, 80%, 90%, 95% or 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, for use in a method of reducing or eliminating the content of triterpenes and/or triterpenoids of latex or rubber from a laticiferous plant or part thereof.

EXAMPLES

Example 1: Materials and Methods

(132) Plant Material and Cultivation Conditions

(133) *T. koksaghyz* wild-type and transgenic plants were grown in standard soil (ED73 Einheitserde, Fröndenberger, Germany). Every four weeks a commercial fertilizer was supplemented according to the manufacturer's recommendations (Hakaphos Plus, Compo GmbH, Munster, Germany). The plants were cultivated in controlled growth chambers or in the greenhouse at 18° C. and 20 klux (high pressure sodium lamp, HPS 600 Watts, Greenbud, enhanced yellow and red spectrum) with a 16-h photoperiod.

(134) Total RNA Extraction and cDNA Synthesis

(135) Latex was harvested from wounded *T. koksaghyz* root in 90 µl REB buffer (Schmidt et al., 2010). Subsequently, total RNA was extracted using the innuPREP RNA Mini Kit (Analytik Jena, Jena, Germany) according to manufacturer's instructions.

(136) Full-length cDNA was synthesized from 500 ng total RNA using PrimeScript RT Master Mix (TaKaRa, Clontech, Saint-Germain-en-Laye, France) according to manufacturer's instructions.

(137) Quantitative RT-PCR

(138) Quantitative RT-PCR analysis was performed as previously described (Putter et al., 2017). The *T. koksaghyz* TkOSC1- and TkOSC-RNAi-plants were 9 to 16 weeks old at the time of expression analysis. All oligonucleotide sequences for expression analysis are listed in Table 1. Primer efficiencies and amplification factors are summarized in Table 2.

(139) TABLE-US-00013 TABLE 1 Sequences of oligonucleotides used. Oligo Sequence (5'43') cloning procedures TkOSC1-RNAi-fwd-NcoI

AAACCATGGGCGGAATTGATCTTATAAGCG (SEQ ID NO: 21) TkOSC1-RNAi-rev-XhoI AAAGTCGAGATCCCAAGCTTGAATCGCAC (SEQ ID NO: 22) TkOSC-RNAi-fwd-NcoI AAACCATGGAACAAGAAAATGGTTCTTGG (SEQ ID NO: 23) TkOSC-RNAi-rev-XhoI AAAGTCGAGTCTCCAAGCGCCCATAGCGG (SEQ ID NO: 24) qRT-PCR TkEF1alpha-fw-realtime CGAGAGATTCGAGAAGGAAGC (SEQ ID NO: 25) TkEF1alpha-rv-realtime CTGTGCAGTAGTACTTGGTGG (SEQ ID NO: 26) TkLUP-fw-

real-time GCTGACCAACCAACAC (SEQ ID NO: 27) TkLUP-rv-realtime
 AGCACGTTCTCTTCGGTTCCAG (SEQ ID NO: 28) TkOSC1-fw-realtime
 ACTCTCCCTTGATAATTGCC (SEQ ID NO: 29) TkOSC1-rv-realtime
 TTGTGCTTCTGCCTGATATAGAAC (SEQ ID NO: 30) TkOSC2-fw-realtime
 CCGGTGAGAAGGTGGAAGTT (SEQ ID NO: 31) TkOSC2-rv-realtime
 GGAACCGGTACCTCCCAAAC (SEQ ID NO: 32) TkOSC3-fw-realtime
 TCCATCCAAACCACAGAAAAG (SEQ ID NO: 33) TkOSC3-rv-realtime
 ATGAAGCATACTCCCCAATAAC (SEQ ID NO: 34) TkOSC4-fw-realtime
 CCGACAATTTCGTAAGGCCACTG (SEQ ID NO: 35) TkOSC4-rv-realtime
 TGTTTGCACCACGTTTCGACC (SEQ ID NO: 36) TkOSC5-fw-realtime
 GAAACACAACCTAGAAGATGGCGGT (SEQ ID NO: 37) TkOSC5-rv-realtime
 CATAGCCCATGAAGTGTGCACT (SEQ ID NO: 38) TkOSC6-fw-realtime
 GGTCATAGCACCATGTTTGGG (SEQ ID NO: 39) TkOSC6-rv-realtime
 GGTGACTGAGCCATGATCCAGG (SEQ ID NO: 40) TkOSC7-fw-realtime
 CGATGCTGGAACACCCGC (SEQ ID NO: 41) TkOSC7-rv-realtime
 GTTGGTGGTATGCTCGATACATC (SEQ ID NO: 42) TkOSC8-fw-realtime
 CTGAGCATACCACCTGCGAG (SEQ ID NO: 43) TkOSC8-rv-realtime
 CAGCTTTTCTTATTGATGTCGTCCTG (SEQ ID NO: 44) TkRP-fw-realtime
 CGTCGATCTCAAGGATGTTGTC (SEQ ID NO: 45) TkRP-rv-realtime
 GGAGCTTTGAGAAGAACCAACG (SEQ ID NO: 46)

(140) TABLE-US-00014 TABLE 2 Primer efficiency and amplification factors of qRT-PCR oligos. All values are obtained from cDNA synthesized from *T. koksaghyz* mRNA. The values were calculated using the Bio-Rad CFX Manager v3.1 software (Bio-Rad Laboratories Inc., Hercules, CA, USA) and the qPCR primer efficiency calculator provided by Thermo Fisher Scientific (thermoscientificbio.com/webtools/qpcr/efficiency/). Oligo pair Efficiency Amplification factor (66° C.) TkLUP-realtime 99.17% 1.99 TkOSC1-realtime 100.16% 2.00 TkOSC2-realtime 101.87% 2.02 TkOSC3-realtime 109.67% 2.10 TkOSC4-realtime 108.54% 2.09 TkOSC5-realtime 106.95% 2.07 TkOSC6-realtime 94.39% 1.94 TkOSC7-realtime 91.2% 1.91 TkOSC8-realtime 98.6% 1.99 TkEF1alpha-realtime 104.48% 2.04 TkRP-realtime 105.44% 2.05 TkSQE1-realtime 95.53% 1.96 TkSQS1-realtime 102.56% 2.03

Cloning and Stable Transformation Procedures

(141) For cloning of RNAi-vectors for stable transformation, cDNA fragments were inserted initially, into the Gateway-compatible entry vector pBluescript II KS (+) (Addgene, Cambridge, MA, USA). Therefore, cDNA fragments TkOSC1-RNAi (125 bp) (SEQ ID NO: 48) and TkOSC-RNAi (445 bp) (SEQ ID NO: 47) were amplified by using the oligos TkOSC1-RNAi-fwd-Nco1 (SEQ ID NO: 21) and TkOSC1-RNAi-rev-Xho1 (SEQ ID NO: 22) or TkOSC-RNAi-fwd-Nco1 (SEQ ID NO: 23) and TkOSC-RNAi-rev-Xho1 (SEQ ID NO: 24), respectively. The insertion of the RNAi-fragments by using the Nco1/Xho1 restriction sites of pBluescript II KS (+) resulted in the entry vectors pBluescript-TkOSC1 and pBluescript-TkOSC. Subsequently, the entry vectors were used for recombination into the expression vector pLab12.5-pREF (Epping et al., 2015) resulting in the expression vectors pLab12.5-pREF-TkOSC1-RNAi (SEQ ID NO: 50) and pLab12.5-pREF-TkOSC-RNAi (SEQ ID NO: 49) vectors. The fragment according to SEQ ID NO: 48 (GCGGAATTGATCTTATAAGCGTACCTCCAGTGAGACTTGGAGACGACGAAGATGTGAACTATGAAGCGGTGACAACAGCCGTGAGAAAGGCAGTTAGACTGAACCGTGCGATTCAAGCTTGGGAT) was used to construct pLab12.5-pREF-TkOSC1-RNAi (SEQ ID NO: 50).

(142) The fragment according to SEQ ID NO: 47

(AACAAGAAAATGGTTCTTGGTATGGTTATTGGG
 GTATATGTTATCTTTATGGAACCTATTTTCGTGTTACAAGGATTGGTAGCTTGTGGACAAACA
 TATGAAAATAGTGAAGCAGTTCGGAAAGCTGTCAAGTTTTTCCTCTCGACACAAAACCTCAGA
 AGGTGGTTGGGGAGAGAACTTTGAGTCATGCCCACAAGAGAAATTTATACCTTTGGAAGGG

AACAGAACAAATTTTGCTGCAAACACTTTCATGGCCATGCTTGGTCTTCTATGTGGTGGACAGGCTGAAAGAGATCCGACACCATTACACAAGGCAGCAAAATTGCTCATAAATGGACAAATGGA TAATGGAGACTTTTCCTCAACAGGAAATAACAGGAGTGACATGAAGAATTGCATGTTACATT ATGCAGAATATAGGAACACTTTTCCGCTATGGGCGCTTGGAGA) was used to construct pLab12.5-pREF-TkOSC-RNAi (SEQ ID NO: 49).

(143) All vector sequences were verified by sequencing. The complete vector sequence or construct for pLab12.5-pREF-TkOSC1-RNAi is given as SEQ ID NO: 50. The complete vector sequence or construct for pLab12.5-pREF-TkOSC-RNAi is given as SEQ ID NO: 49. Oligonucleotide sequences are shown in Table 1.

(144) The transformation of *T. koksaghyz* by *A. tumefaciens* strain EHA105 with OE- and RNAi-constructs was carried out as previously described (Stolze et al., 2017).

(145) Triterpene Extraction and Analysis

(146) Whole *T. koksaghyz* roots were frozen in liquid nitrogen, freeze-dried, and grinded to a fine powder. Root material was harvested from 5- to 7-month old RNAi-lines. For triterpene analysis, 100 mg of dry root material were used for saponification by adding 20 ml of methanol containing 6% potassium hydroxide and heating to 80° C. for 2 h. 100 µl of betuline (2.5 mg/ml stock solutions in acetone) were added as internal standard. Samples were refilled with water to a volume of 15 ml and extracted three times with one volume of hexane by vortexing and subsequent centrifugation at 4000 rpm for 5 min. The pooled hexane phases were evaporated in the Rocket Evaporator 4D (Genvac Aerospace Inc., Cleveland, USA) and redissolved in 1 ml acetone.

(147) To extract NR from root material, 2 g of the dry root powder as mentioned above and 20 ml of water were transferred into a 50 ml reaction tube and a mixture of steal beads (six 7 g-beads and nine 4 g-beads) were added. The sample was vortexed for 30 min and centrifuged for 15 min at 5000 g. The floating rubber flakes were removed, washed in fresh water and dried over night at 40° C. For triterpene extraction, 100 mg of the resulting NR were filled up with 10 ml acetone and 100 µl internal betuline standard (2.5 mg/ml stock solution in acetone) and were incubated by rotating at room temperature for one week. The resulting extract was dried and redissolved in 1 ml acetone.

(148) Triterpene analysis was performed with a GC-MS-QP 210 Ultra system (Shimadzu, Duisburg, Germany) equipped with a Rxi®-5 ms column (Restek GmbH, Bad Homburg, Germany). 0.5 µl of the extract were injected using split modus (1:10) at an injector and interface temperature of 260° C. The GC temperature program was as follows: 120° C. for 3 min, temperature gradient of 15° C. per minute up to 330° C., 330° C. for 10 min. Electron ionization (EI) in the MS was set to 70 eV. Peak integration and identification was performed with the LabSolution software (Shimadzu, Duisburg, Germany) using a NIST library (NIST=National Institute of Standards and Technology) or by analyzing the corresponding standards obtained from Extrasynthese (Genay, France). Quantification was performed in relation to the internal standard. Retention indices for the triterpene compounds were determined using the LabSolution software in relation to a C8-C40 alkane calibration standard (Sigma Aldrich, Taufkirchen, Germany) using the same operating conditions.

(149) Statistical Analysis

(150) For statistical analysis with more than three samples, the Mann-Whitney U test was applied. This test is a nonparametric test of the null hypothesis that does not require the assumption of normal distributions. The corresponding results were displayed in boxplots. All other statistics were performed using the two-tailed t-test. If variances were assumed to be unequal Welch's correction was applied. The corresponding results were displayed in bar charts.

Example 2: Generation of *T. koksaghyz* Plants Comprising RNAi Targeting OSCs

(151) In order to gain *T. koksaghyz* plants with reduced triterpene levels in latex, two different RNAi constructs driven by the laticifer predominant REF-promoter were transformed in *T. koksaghyz* plants. The fragment (SEQ ID NO: 48) used to construct plab12.5-pREF-TkOSC1-RNAi (SEQ ID NO: 50) (comprising SEQ ID NO: 2 as the RNAi that is specifically directed

against the mRNA transcript) comprises a 125 bp long not conserved region and was supposed to knock-down only TkOSC1, whereas the fragment (SEQ ID NO: 47) of pLab12.5-pREF-TkOSC-RNAi (SEQ ID NO: 49) (comprising SEQ ID NO: 1 as the RNAi that is specifically directed against the mRNA transcript) contains a 445 bp long sequence derived from a more conserved region and was expected to knock-down several TkOSCs. After stable transformation, the inventors received two individual TkOSC1-RNAi lines comprising seven plants and five TkOSC-RNAi lines consisting of 12 plants.

(152) In 9- to 16-week-old plants, the expression of TkOSC1 in latex was analysed via qPCR (FIG. 1(a)). Compared to *T. koksaghyz* wild-type plants (TkWt; n=4) the expression of TkOSC1 was reduced significantly in all lines. For the TkOSC1-RNAi plants, the expression was reduced to one-seventh. For the TkOSC-RNAi plants, the effect was even stronger, resulting in a reduction to one-eighteenth.

(153) To analyse the expression of further triterpene biosynthesis related genes, three wild-type, three TkOSC1-RNAi plants and four TkOSC-RNAi plants were exemplarily used for further qPCR analysis (FIG. 1(b) and FIG. 1(c)). TkOSC-RNAi plants show a remarkable reduction of TkOSC2, TkOSC3, TkOSC5 and TkLUP expression. Only TkOSC6 expression was slightly enhanced. Surprisingly, the plants with the TkOSC1 specific construct showed also a reduced expression of TkOSC2, TkOSC3 and TkOSC5, but the effect was weaker compared to the TkOSC-RNAi plants. TkOSC6 expression was not affected and TkLUP expression was slightly upregulated. TkOSC4 expression was not detectable in latex of any plants. Earlier studies revealed the co-regulation of TkOSC1 with the upstream genes TkSQS1 and TkSQE1 (Unland et al., 2018). Therefore, the expression of TkSQS1 and TkSQE1 was also analysed in the RNAi-lines (FIG. 3(c)). Mean expression of TkSQS1 in TkOSC1-RNAi plants was reduced to one-half, whereas the TkSQE1 expression was comparable to the expression in wild-type plants. In the TkOSC-RNAi lines, TkSQS1 expression was even further reduced to one-quarter and TkSQE1 expression was reduced to one-half compared to the wild-type, confirming the co-regulation of TkSQS1, TkSQE1 and TkOSC1. It can be noted that the expression of several triterpene biosynthesis related genes are affected in both RNAi lines with a general stronger effect in the plants carrying the TkOSC-RNAi fragment.

Example 3: Effect of Oxidosqualene Cyclase (OSC) Knockdown on Triterpene Composition

(154) To analyse if alterations in expression have an effect on triterpene composition in the RNAi lines, triterpene extracts of dry root material from 5- to 7-month-old plants were measured by GC/MS (FIG. 1). The pentacyclic triterpenes were strongly affected in the transgenic lines (FIG. 1(d) and FIG. 1(e)). As a result, the total amount was reduced from 4.6 mg/g in wild-type plants to 2.8 mg/g in TkOSC1-RNAi plants and to 2.6 mg/g in TkOSC-RNAi plants. Especially the most abundant triterpenes, taraxasterol, β -amyrin and one unidentified triterpene (n.i.1), were remarkably reduced with a higher level of significance in TkOSC-RNAi plants.

(155) Besides the pentacyclic triterpenes, further compounds were measured via GC/MS (FIG. 1(e), Table 3). It was found that precursors like squalene and 2,3-oxidosqualene accumulated significantly in the transgenic lines, with a higher level of significance in TkOSC-RNAi plants. Additionally, cycloartenol and 24-methylene cycloartanol, two sterol intermediates, were detected in both transgenic lines, with an average of 0.1 mg/g dry root weight in TkOSC1-RNAi plants and 0.9 mg/g dry root weight in TkOSC-RNAi plants. Sterols including campesterol, stigmasterol and sitosterol were detected to equal amounts of about 0.5 mg/g dry root weight in all plants and were not affected by the knock-down. In sum, the total triterpene content including precursors, sterol intermediates, sterols and pentacyclic triterpenes was significantly reduced in transgenic RNAi lines to about two-thirds with a higher level of significance in TkOSC-RNAi plants (FIG. 1(e)).

(156) TABLE-US-00015 TABLE 3 Triterpene content of root material from *T. koksaghyz* TkOSC1-/TkOSC-RNAi lines. Triterpene compounds were identified and quantified by GC/MS [mg/g root dry weight], corresponding retention indices (RI) were determined in relation to a C8-

C40 alkane calibration standard. The mean values represent independent extractions from wild-type plants (n = 4), TkOSC1-RNAi plants (n = 7) and TkOSC-RNAi plants (n = 12) conducted in triplicates and are indicated with the corresponding standard deviations. RI TkWt TkOSC1-RNAi TkOSC-RNAi precursors 0.05 ± 0.01 0.08 ± 0.04 0.17 ± 0.08 squalene 2843 0.01 ± 0.00 0.01 ± 0.00 0.03 ± 0.02 2,3-oxidosqualene 2965 0.04 ± 0.01 0.06 ± 0.03 0.14 ± 0.08 sterol intermediates 0.00 ± 0.00 0.14 ± 0.06 0.27 ± 0.07 cycloartenol 3478 0.00 ± 0.00 0.09 ± 0.03 0.15 ± 0.04 24-methylene cycloartanol 3536 0.00 ± 0.00 0.05 ± 0.04 0.13 ± 0.03 sterols 0.50 ± 0.06 0.52 ± 0.05 0.54 ± 0.05 campesterol 3313 0.07 ± 0.01 0.07 ± 0.01 0.07 ± 0.01 stigmasterol 3343 0.22 ± 0.04 0.21 ± 0.02 0.24 ± 0.03 sitosterol 3405 0.21 ± 0.01 0.24 ± 0.03 0.23 ± 0.02 pentacyclic triterpenes 4.57 ± 0.38 2.79 ± 0.58 2.60 ± 0.42 taraxerol and n.i. 3448 0.70 ± 0.09 0.81 ± 0.14 0.79 ± 0.10 β-amyrin 3465 0.73 ± 0.06 0.25 ± 0.08 0.23 ± 0.05 lupeol 3510 0.12 ± 0.02 0.18 ± 0.07 0.11 ± 0.02 α-amyrin and lup-19(21)-en-3-ol 3516 0.70 ± 0.06 0.52 ± 0.14 0.56 ± 0.17 TkOSC3-product-1 3550 0.06 ± 0.01 0.05 ± 0.01 0.07 ± 0.01 TkOSC3-product-2 3590 0.48 ± 0.05 0.45 ± 0.10 0.58 ± 0.12 n.i.1 3603 0.43 ± 0.04 0.19 ± 0.06 0.12 ± 0.02 taraxasterol 3615 1.37 ± 0.12 0.33 ± 0.13 0.14 ± 0.05

(157) For triterpene quantification in NR, root material of three wild-type, TkOSC1-RNAi and TkOSC-RNAi plants, respectively, were used for rubber extraction. The triterpenes in NR extracts were isolated by acetone extraction and quantified in relation to NR weight (FIG. 2, Table 4). The crude acetone extracts contained acetate derivatives of different triterpenes. Those acetates as well as the predominant triterpenes in latex (taraxasterol, β-amyrin and n.i.1) were strongly reduced compared to wild-type plants with a higher level of significance in TkOSC-RNAi plants compared to TkOSC1-RNAi plants (FIGS. 2(a) and 2(b)). The average amount of total pentacyclic triterpenes in wild-type plants with 56.6 mg/g NR was reduced to 33.4 mg/g NR in TkOSC1-RNAi lines or even to 16.1 mg/g NR in TkOSC-RNAi lines (FIG. 2(c)).

(158) TABLE-US-00016 TABLE 4 Triterpene content in NR extracts from *T. koksaghyz* TkOSC1-/TkOSC-RNAi lines. Triterpene compounds were identified and quantified by GC/MS [mg/g NR dry weight], corresponding retention indices (RI) were determined in relation to a C8-C40 alkane calibration standard. The mean values represent independent extractions from wild-type plants (WtTk; n = 3), TkOSC1-RNAi plants (n = 3) and TkOSC-RNAi plants (n = 3) and are indicated with the corresponding standard deviations. RI TkWt TkOSC1-RNAi TkOSC-RNAi precursors 1.02 ± 0.69 1.69 ± 0.92 5.12 ± 1.14 squalene 2843 0.59 ± 0.53 0.54 ± 0.39 0.58 ± 0.23 2,3-oxidosqualene 2965 0.44 ± 0.17 1.16 ± 0.53 4.55 ± 0.95 sterol precursors 0.00 ± 0.00 1.86 ± 0.23 3.51 ± 0.06 cycloartenol 3478 0.00 ± 0.00 1.48 ± 0.46 2.06 ± 0.11 24-methylene cycloartanol 3536 0.00 ± 0.00 0.37 ± 0.30 1.45 ± 0.06 sterols 1.25 ± 0.15 2.14 ± 0.63 1.45 ± 0.09 campesterol 3313 0.15 ± 0.08 0.23 ± 0.07 0.16 ± 0.02 stigmasterol 3343 0.43 ± 0.04 0.59 ± 0.19 0.51 ± 0.05 sitosterol 3405 0.67 ± 0.14 1.32 ± 0.38 0.78 ± 0.02 pentacyclic triterpenes 56.60 ± 0.62 33.37 ± 3.99 16.05 ± 2.86 taraxerol and n.i. 3448 1.84 ± 0.14 3.65 ± 0.85 1.93 ± 0.21 β-amyrin 3465 6.14 ± 0.60 3.69 ± 0.95 1.83 ± 0.32 lupeol 3510 1.85 ± 0.91 3.53 ± 0.29 1.30 ± 0.24 α-amyrin and lup-19(21)-en-3-ol 3516 4.74 ± 0.34 7.89 ± 2.03 6.46 ± 1.40 β-amyrin acetate 3562 3.17 ± 0.47 0.47 ± 0.27 0.00 ± 0.00 TkOSC3-product-1 3550 0.08 ± 0.01 0.28 ± 0.12 0.17 ± 0.02 TkOSC3-product-2 3590 0.62 ± 0.06 1.49 ± 0.43 1.43 ± 0.24 n.i.1 3603 8.27 ± 0.07 4.54 ± 0.97 1.09 ± 0.26 taraxasterol 3615 22.58 ± 1.68 7.00 ± 0.77 1.85 ± 0.59 triterpene acetate 1 3695 2.41 ± 0.40 0.54 ± 0.20 0.00 ± 0.00 triterpene acetate 2 3707 4.90 ± 1.14 0.33 ± 0.24 0.00 ± 0.00

(159) In contrast to the reduced pentacyclic triterpene content, increasing amounts of triterpene precursors as well as sterol intermediates were again detected in RNAi-lines compared to wild-type, while equal amounts of sterols were measured in all three groups. However, total triterpenoid amount was still significantly decreased to two thirds in TkOSC1-RNAi NR and to less than the half in TkOSC-RNAi NR (FIG. 2(b)).

(160) In conclusion, it can be claimed that an OSC knockdown strongly reduces triterpenes/triterpenoids in dandelion root and NR.

(161) The listing or discussion of a previously published document in this specification should not necessarily be taken as an acknowledgement that the document is part of the state of the art or is common general knowledge.

(162) The invention illustratively described herein may suitably be practiced in the absence of any element or elements, limitation or limitations, not specifically disclosed herein. Thus, for example, the terms “comprising”, “including”, “containing”, etc. shall be read expansively and without limitation. Additionally, the terms and expressions employed herein have been used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention claimed. Thus, it should be understood that although the present invention has been specifically disclosed by exemplary embodiments and optional features, modifications and variations of the inventions disclosed herein are also considered to be within the scope of this invention.

(163) The invention has been described broadly and generically herein. Each of the narrower species and sub-generic groupings falling within the generic disclosure also form part of the invention. This includes the generic description of the invention with a proviso or negative limitation removing any subject matter from the genus, regardless of whether or not the excised material is specifically recited herein.

(164) Other embodiments are within the following claims. In addition, where features or aspects of the invention are described in terms of Markush groups, those skilled in the art will recognize that the invention is also thereby described in terms of any individual member or subgroup of members of the Markush group.

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Claims

1. A method of reducing the content of triterpenes, triterpenoids, or both triterpenes and triterpenoids, of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to a like laticiferous plant or part thereof that has not been genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase, wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase, wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing the activity of the at least one oxidosqualene cyclase, wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase and thereby affects the presence of the protein of the at least one oxidosqualene cyclase, wherein the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and wherein the at least one RNAi comprises or consists of a nucleotide sequence being at least 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, wherein the at least one RNAi attaches individually or binds to the mRNA transcript of the at least one oxidosqualene cyclase, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being at least 99% identical to SEQ ID NO: 4.

2. The method according to claim 1, wherein the laticiferous plant or part thereof is selected from the group of plant families consisting of Apocynaceae, Asteraceae, Euphorbiaceae and Moraceae.

3. The method according to claim 1, wherein the laticiferous plant or part thereof is selected from the group consisting of *Agoseris glauca*, *Apocynum venetum*, *Chrysothamnus graveolens*, *Chrysothamnus nauseosus*, *Chrysothamnus viscidiflorus*, *Funtumia elastica*, *Parthenium argentatum*, *Scorzonera acanthoclada*, *Scorzonera albicaulis*, *Scorzonera divaricate*, *Scorzonera hissaricata*, *Scorzonera racemosa*, *Scorzonera tau-saghyz*, *Scorzonera tragapogonoides*, *Scorzonera turkestanica*, *Scorzonera virgate*, *Solidago canadensis scabra*, *Solidago fistulosa*, *Solidago leavenworthii*, *Solidago rigida*, *Sonchus oleraceus*, *Taraxacum brevicorniculatum*, *Taraxacum hybernum*, *Taraxacum koksaghyz*, *Taraxacum megalorrhizon*, *Taraxacum officinale* and crossings thereof.

4. The method according to claim 1, wherein the at least one oxidosqualene cyclase is selected from the group consisting of oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to

SEQ ID NO: 8, the oxidosqualene cyclase 4 according to SEQ ID NO: 10, the oxidosqualene cyclase 5 according to SEQ ID NO: 12, the oxidosqualene cyclase 6 according to SEQ ID NO: 14, the oxidosqualene cyclase 7 according to SEQ ID NO: 16, the oxidosqualene cyclase 8 according to SEQ ID NO: 18, and the lupeol synthase according to SEQ ID NO: 20.

5. The method according to claim 1, wherein the part of the laticiferous plant is selected from the group consisting of cells, tissues, organs, roots, stems, branches, leaves, exudates, latex and mixtures thereof.

6. The method according to claim 1, wherein the at least one RNAi comprises or consists of the nucleotide sequence according to SEQ ID NO: 1, or comprises or consists of the nucleotide sequence according to SEQ ID NO: 2.

7. The method according to claim 3, wherein the laticiferous plant or part thereof is selected from the group consisting of *Parthenium argentatum*, *Chrysanthamnus nauseosus*, *Taraxacum koksaghyz*, *Taraxacum officinale* and crossings thereof.

8. The method according to claim 4, wherein the at least one oxidosqualene cyclase is selected from the group consisting of the oxidosqualene cyclase 1 according to SEQ ID NO: 4, the oxidosqualene cyclase 2 according to SEQ ID NO: 6, the oxidosqualene cyclase 3 according to SEQ ID NO: 8, and the oxidosqualene cyclase 5 according to SEQ ID NO: 12, and an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8 or 12.

9. The method according to claim 1, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being identical to SEQ ID NO: 4.

10. The method according to claim 1, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, or 20.

11. The method according to claim 1, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being identical to SEQ ID NO: 4.

12. The method according to claim 1, wherein taraxasterol and/or β -amyrin, is/are reduced in the laticiferous plant or part thereof.

13. The method according to claim 1, wherein the plant is (a) *Taraxacum koksaghyz*, (b) wherein the part of the plant is the plant root, or both (a) and (b).

14. A method of increasing the content of a triterpene precursor, a triterpenoid precursor, or both triterpene and triterpenoid precursors, of latex or rubber from a laticiferous plant or part thereof, comprising genetically modifying the laticiferous plant or part thereof, such that the activity of at least one oxidosqualene cyclase is reduced, compared to a like laticiferous plant or part thereof that has not been genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase, wherein the laticiferous plant or part thereof is genetically modified by at least one heterologous nucleotide sequence that is capable of reducing the activity of the at least one oxidosqualene cyclase, wherein the at least one heterologous nucleotide sequence encodes at least one RNAi that interferes with the mRNA encoding the at least one oxidosqualene cyclase, with the proviso that the at least one RNAi is capable of reducing the activity of the at least one oxidosqualene cyclase, wherein the at least one RNAi is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase and thereby affects the presence of the protein of the at least one oxidosqualene cyclase, wherein

the at least one oxidosqualene cyclase is selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and wherein the at least one RNAi comprises or consists of the nucleotide sequence being at least 99% identical to SEQ ID NO: 1 or SEQ ID NO: 2, wherein the at least one RNAi attaches individually or binds to the mRNA transcript of the at least one oxidosqualene cyclase, wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 1 is specifically directed against the mRNA transcript of the at least one oxidosqualene cyclase selected from the group consisting of an amino acid sequence being at least 99% identical to any one of SEQ ID NOs: 4, 6, 8, 10, 12, 14, 16, 18, or 20, and wherein the at least one RNAi comprising or consisting of a nucleotide sequence being at least 99% identical to SEQ ID NO: 2 is specifically directed against the mRNA transcript of the oxidosqualene cyclase of an amino acid sequence being at least 99% identical to SEQ ID NO: 4.

15. The method according to claim 14, wherein one or more of 2,3-oxidosqualene, cycloartenol, 24-methylene cycloartenol is increased in the lactiferous plant or part thereof.
